MIL-C-3965/22B
26 August 1970
SUPERSEDING
MIL-C-3965/22A
29 October 1968

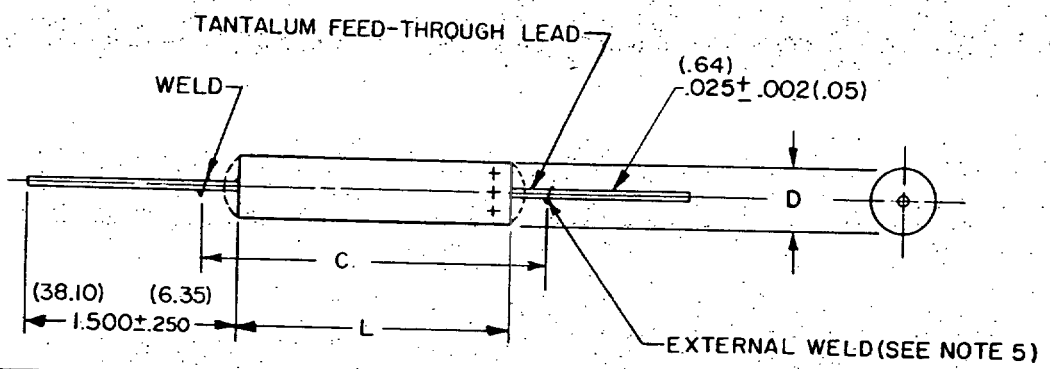
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, HIGH ETCHED FOIL) - 85°C (VOLTAGE DERATED TO 125°C), STYLES CL70 AND CL71

INACTIVE FOR NEW EQUIPMENT DESIGN AFTER DATE OF THIS SPECIFICATION; USE MIL-C-39006/16

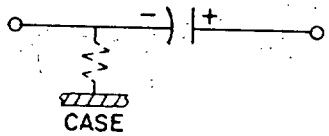
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



Case size	Dimensions			
	1/ L ±.062 (1.57)	D ±.016 (.41)	D Max	C Max
		CL70	CL71	
E1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)
E2	.906 (23.01)	.281 (7.14)	.312 (7.92)	1.406 (35.71)
E3	1.438 (36.53)	.375 (9.53)	.406 (10.31)	1.938 (49.23)
E4	2.125 (53.98)	.375 (9.53)	.406 (10.31)	2.625 (66.68)
E5	2.750 (69.85)	.375 (9.53)	.406 (10.31)	3.250 (82.55)

1/ For style CL71, nonshrinkable sleeving shall extend .016 inch minimum, .062 inch maximum, beyond each end of the case; shrinkable sleeving shall lap over the ends of the case.



CIRCUIT DIAGRAM
(SEE NOTE 4)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. There is an indeterminate resistance between the metal case and the negative terminal.
5. The weld shall not be enclosed in the end seal.

FIGURE 1. Styles CL70 and CL71 capacitors.

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TABLE I. Capacitor characteristics.

Type designation 1/	Voltage			Capacitance	DC leakage (max)			Impedance (max)	Case size
	Rated (85° C)	Derated (125° C)	Surge (85° C)		25° C	85° C	125° C		
	volts, dc	volts, dc	volts, dc	μF	μA	μA	μA	ohms	
CL7-BE290UPE	15	10	17.2	29	7	35	23	137	E1
CL7-BE101UPE	15	10	17.2	100	23	100	80	37	E2
CL7-BE311UPE	15	10	17.2	310	61	370	250	10.8	E3
CL7-BE651UPE	15	10	17.2	650	130	780	520	5.1	E4
CL7-BE921UPE	15	10	17.2	920	185	1100	750	4.0	E5
CL7-BG170UPE	25	15	28.8	17	6	34	21	220	E1
CL7-BG600UPE	25	15	28.8	60	23	100	72	62	E2
CL7-BG181UPE	25	15	28.8	180	60	360	220	20.5	E3
CL7-BG381UPE	25	15	28.8	380	125	760	460	9.6	E4
CL7-BG551UPE	25	15	28.8	550	185	1100	660	6.7	E5
CL7-BH150UPE	30	20	34.5	15	6	34	22	250	E1
CL7-BH520UPE	30	20	34.5	52	23	120	80	71	E2
CL7-BH151UPE	30	20	34.5	150	60	360	240	25	E3
CL7-BH321UPE	30	20	34.5	320	125	750	500	11.5	E4
CL7-BH461UPE	30	20	34.5	460	165	1000	700	8	E5
CL7-BJ070TPE	50	30	57.5	7	5	28	17	550	E1
CL7-BJ250TPE	50	30	57.5	25	19	100	60	150	E2
CL7-BJ750TPE	50	30	57.5	75	50	300	180	49	E3
CL7-BJ161TPE	50	30	57.5	160	100	640	390	23	E4
CL7-BJ231TPE	50	30	57.5	230	150	920	550	16	E5
CL7-BL040TPE	75	50	86.2	4	5	24	16	930	E1
CL7-BL150TPE	75	50	86.2	15	17	90	60	250	E2
CL7-BL450TPE	75	50	86.2	45	45	270	180	83	E3
CL7-BL950TPE	75	50	86.2	95	95	570	380	39	E4
CL7-BL131TPE	75	50	86.2	130	130	780	520	29	E5
CL7-BN030SPE	100	65	115.0	3	4	20	14	1210	E1
CL7-BN100SPE	100	65	115.0	10	15	80	52	370	E2
CL7-BN300SPE	100	65	115.0	30	40	230	150	121	E3
CL7-BN630SPE	100	65	115.0	63	80	480	310	58	E4
CL7-BN900SPE	100	65	115.0	90	120	720	470	41	E5
CL7-BQ1R6SPE	150	100	172.0	1.6	3	18	12	2340	E1
CL7-BQ5R5SPE	150	100	172.0	5.5	12	64	42	675	E2
CL7-BQ170SPE	150	100	172.0	17	30	190	130	220	E3
CL7-BQ350SPE	150	100	172.0	35	65	400	260	106	E4
CL7-BQ500SPE	150	100	172.0	50	95	560	380	74	E5

1/ Complete type designation will include an additional symbol to indicate style.

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TABLE II. Cross-reference data.

FROM Type designation 1/ MIL-C-3965/22B	TO Part number 2/ MIL-C-39006/16	FROM Type designation 1/ MIL-C-3965/22B	TO Part number 2/ MIL-C-39006/16
	M39006/16-		M39006/16-
CL7-BE290UPE	1000	CL7-BJ161TPE	1018
CL7-BE101UPE	1001	CL7-BJ231TPE	1019
CL7-BE311UPE	1002	CL7-BL040TPE	1020
CL7-BE651UPE	1003	CL7-BL150TPE	1021
CL7-BE921UPE	1004	CL7-BL450TPE	1022
CL7-BG170UPE	1005	CL7-BL950TPE	1023
CL7-BG600UPE	1006	CL7-BL131TPE	1024
CL7-BG181UPE	1007	CL7-BN030SPE	1025
CL7-BG381UPE	1008	CL7-BN100SPE	1026
CL7-BG551UPE	1009	CL7-BN300SPE	1027
CL7-BH150UPE	1010	CL7-BN630SPE	1028
CL7-BH520UPE	1011	CL7-BN900SPE	1029
CL7-BH151UPE	1012	CL7-BQ1R6SPE	1030
CL7-BH321UPE	1013	CL7-BQ5R5SPE	1031
CL7-BH461UPE	1014	CL7-BQ170SPE	1032
CL7-BJ070TPE	1015	CL7-BQ350SPE	1033
CL7-BJ250TPE	1016	CL7-BQ500SPE	1034
CL7-BJ750TPE	1017		

1/ Complete type designation will include an additional symbol to indicate style.
2/ Part number referenced will replace styles CL70 and CL71.

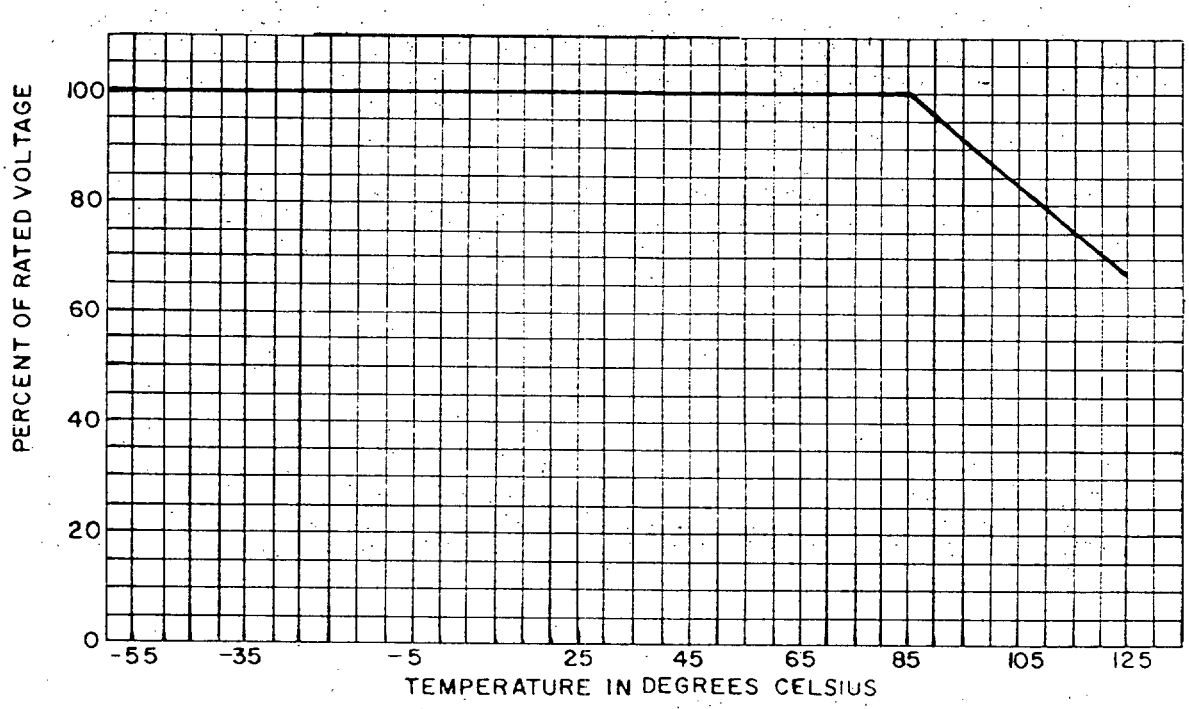


FIGURE 2. Voltage derating with temperature.

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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Tubular; uninsulated (CL70), insulated (CL71).
Seal type - Nonhermetic (symbol E).
Terminals - Axial wire-lead.
Rated temperature - -55° to +85° C, voltage derated to +125° C (see figure 2).
Rated voltage - See table I.
Capacitance (cap) (nom) - See table I.
Capacitance tolerance - Symbol S (+30, -15%), T (+50, -15%), or U (+75, -15%) as shown in table I.

DC leakage (DCL) (max) (at +25° C):
See table I.

Dissipation factor (DF) (max):
20% (≤50 v) or 15% (>50 v).

Seal: Not applicable.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - 2,000 vdc.
Terminal to case - Not applicable.

Insulation resistance:
Sleeving to case (min): 100 megohms.
Terminal to case - Not applicable.

Temperature cycling:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - 115% of table I value.

Solderability: In accordance with MIL-C-3965.

Terminal strength:
Pull test - In accordance with MIL-C-3965.
Bend test - Not applicable.
Torque test - Not applicable.
Wire-lead bend test - In accordance with MIL-C-3965.
Twist test - Not applicable.

Surge voltage: See table I.

Moisture resistance:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±6% of initial measurement.
DF (max) - 115% of table I value.

Low temperature (storage):
DCL (max) - See table I (25° C).
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Stability at low and high temperature:
Step 1 (+25° C):
DCL (max) - See table I.
Cap - Within tolerance of table I value.
DF (max) - See table I.
Step 2 (-55° C):
Impedance - See table I.
ΔCap (max) - -40% of step 1 value.
Step 3 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85° C):
DCL (max) - See table I.
ΔCap (max) - +50% of step 1 value.
DF (max) - 55% (≤50 v) or 45% (>50 v).
1/ Step 5 (+125° C):
DCL (max) - See table I.
ΔCap (max) - +75% of step 1 value.
DF (max) - 55% (≤30 v) or 45% (>30 v).
Step 6 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Reverse voltage:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±10% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Life at +85° C:
2,000-hr (qualification):
DCL (max) - See table I (25° C).
ΔCap - Within ±15% of initial measurement.
DF (max) - Shall not exceed initial requirement.
240-hr (group B):
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - Shall not exceed initial requirement.
1,760-hr (group C):
DCL (max) - See table I (25° C).
ΔCap - Within ±15% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Barometric pressure:
Test voltage - Rated voltage.

1/ 125° C derated voltage referenced.

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Life at +125° C:
DCL (max) - See table I (25° C).
ΔCap - Within ±15% of initial measurement.
DF (max) - Shall not exceed initial requirement.
DWV (min) (CL71, only) - 2,000 vdc.
IR (min) (CL71, only) - 100 megohms.

Marking:
Polarity - Series of plus (+) symbols encircling positive end (see figure 1).

Voltage group (85° C rated voltage):
I - - - - - - - - - - - 15 to 150 v, incl.

NOTE: The capacitors covered by this specification sheet are substitutable for those covered by MIL-C-3965/22A. See table II.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MU
Navy - EC, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Air Force - 80

Agent:
DSA - ES

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